RESEARCH DECISION / 13 OCTOBER 2026

Should QH use a deeper mathematical image pipeline?

Recommendation: extend the current deterministic rig with an optional analysis and deformation stage. Keep the stable character origin, joint hierarchy and time-addressable motion. Add semantic masks, hidden-part completion, joint fitting and local mesh deformation during asset preparation. Evaluate depth and shading as optional signals.

The proposed direction is scientifically sound. A literal grayscale conversion is insufficient: the useful output is a collection of numeric maps with different meanings. Image interpretation and motion generation remain distinct problems. Mathematical constraints make motion controllable; they cannot reveal every hidden surface from a single picture.

Approach	Main benefit	Main limitation	Decision
Current raster rig	Stable origin; editable joint transforms; deterministic playback	Authored pivots and overlap art; rigid clothing bends	Keep as the runtime foundation
Full automatic perception + depth + shadows	Can reduce preparation of unfamiliar, complex artwork	Uncertain hidden anatomy, ordering and stylized shading; model cost	Research as an optional preparation route
Hybrid semantic layers + constrained mesh	Adds natural bends and stronger checks while retaining control	Needs a verified bind pose and part-specific weights	Best candidate for QH

What was actually researched

Reviewed primary papers and author-maintained implementations spanning segmentation, layer decomposition, illustration pose estimation, depth, shading, registration, deformation, tracking, temporal smoothness and compositing. The reference list records the evidence level. No large vision model was installed or benchmarked on Fox. This report is an engineering recommendation, not an empirical claim that a new model beats our rig.

Current project status

The shipped Fox uses a stationary raster master plus sleeve/palm rotations. The separate local studio draft has 27 part controls and matrix-based playback. Both already use mathematics. The gaps are automated semantic extraction, verified hidden artwork, flexible deformation, and a quality measure that reflects those failures. Studio and skill implementation were paused for this research.

Use separate numeric maps

An RGBA image is already a numeric array. Useful preprocessing preserves that original and computes additional channels; it does not replace the original with one filtered image. A mask can be displayed in black and white, but the display colors are only a visualization of its stored numbers.



Illustration from our draft bind pose. Part IDs, draw order and joints are authored data rendered for explanation, not predictions from a vision model. Signed distance is computed from the alpha silhouette.

Field	Meaning	Suggested storage
RGB + alpha	Original color and edge coverage	RGBA PNG; linear floats while calculating
Semantic ID / part probability	Visible part ID plus per-layer alpha; overlaps may contain several parts	uint16 IDs + separate alpha/confidence maps
Signed distance / boundary	Inside/outside distance; smooth seam and contour losses	float map, in pixel units
Relative order / occlusion	Which fragment is in front at an overlap	Ordinal map or constrained graph
Joint + skinning weight	Where parts rotate; how vertices follow bones	Joint JSON + sparse vertex weights
Shading / material / confidence	Optional separation of painted color and illumination	Separate maps with provenance

The closest research to your idea

See-through, SIGGRAPH 2026 [1]: jointly predicts completed RGBA body-part layers and pseudo-depth for drawing order. Its paper reports 19 semantic categories; the current repository offers up to 23 layers. It is the closest match. Published examples concern anime artwork, so success on our Fox is unproven. The authors still report ordering and overlap defects. Their repository explicitly separates decomposition from rigging. [1,2]

Qwen-Image-Layered [3]: provides general RGBA layer decomposition and independent edits. Useful as a baseline for graphic assets. A clean reconstruction alone does not establish that its layers have suitable shoulder, wrist or facial boundaries. This is a QH evaluation requirement, not a measured result for Fox.

SAM 3 / SAM 2 / Semantic-SAM [4-6]: useful for visible-region proposals and interactive correction. SAM 3 accepts concepts; SAM 2 propagates masks through video; Semantic-SAM studies granularity and recognition. A mask proposal must still be assigned a consistent QH part ID and validated. None of these operations alone supplies unseen anatomy or a rig.

Illustration-specific pose work [7,8]: supports adapting joint estimation to drawings rather than trusting human-photo models on a mascot. Animated Drawings combines detection, segmentation, pose and mesh animation. Fox’s ears, tail, shoes and large head need their own anatomy template and joint rules.

Generative Cutout Animation, 2026 [23]: starts from an animated skeleton and a prompt, then optimizes reusable cutouts across target poses. This offers a useful alternative: define motion constraints before generating layers. It is not an automatic recovery of the true anatomy of an existing image.

Neural Puppet [9]: combines a layered 2.5D template with deformation learned from example poses. It supports improving correspondence across images while retaining structure; it does not imply that a single RGB image uniquely determines a complete rig.

Cost belongs in preparation, not every preview frame

See-through reports about 74 seconds for decomposition and 10 seconds for depth on an RTX 4090. Its maintained setup describes roughly 12-16 GB VRAM at 1280 resolution and lower-memory variants. These are upstream figures, not measurements on this Mac; the current two-stage workflow can take longer. Cache accepted layers and geometry. [1,2]

Candidate order for a later experiment

First compare authored layers, prompted visible masks plus targeted completion, and See-through on identical Fox poses. Use Qwen as a general decomposition baseline. Introduce depth and shading separately so we can measure whether each actually improves the output.

Where an arm should rotate

There are several origins: the character root, each joint, each local image buffer, and the display viewport. A single center is insufficient for a body with articulated arms. The character root should be stable; the wrist and elbow should follow the parent chain. Display scaling happens after the rig math.

Keep the body origin independent of moving hands

```
q = s R p + t
E_register = sum_i w_i ||s R p_i + t - q_i||^2
```

Use body-owned correspondences to fit rotation, translation and one scale [10]. Weights and robust outlier rejection are proposed extensions. Full-silhouette centroids can move when the hand or tail moves: that is legitimate silhouette motion, not a reason to recenter the body. Retain scale units and the bind pose in metadata.

Hierarchy and pivot math

```
F_i(t) = F_parent(t) T(a_i) R(theta_i(t)) S_i(t)
p_world = F_i(t) T(-c_i) p_local
```

Here a_i is the attachment offset in the parent joint frame, c_i is the pivot inside the local part image, and F_i maps from that joint frame to world coordinates. A child inherits F_i, not an already shifted image buffer. This avoids applying the pivot correction twice. Keep fractions of a pixel through calculation.

Estimating a pivot from motion observations

```
p' = R(p-c) + c + d
(I-R)c = p' - R p - d
```

With multiple corresponding points and a known rotation/translation, solve for c by weighted least squares. Near zero rotation, I-R is nearly singular; the pivot cannot be located reliably. Translation and center are also confounded if both are unconstrained. A single still image needs anatomical priors or an authored pivot. A contour skeleton is a useful proposal, but clothing pockets and tails can create misleading branches.

Hand targets and reach constraints

```
cos(theta_elbow) = (r^2 - L1^2 - L2^2)/(2 L1 L2)
```

For a two-bone planar arm, clamp reachable distances to |L1-L2| ... L1+L2, choose a consistent elbow direction, and solve the shoulder angle from the target. A wrist follows the forearm; it must not be moved independently in world coordinates while also inheriting the forearm. Use inverse kinematics for contact and forward kinematics for authored waving arcs.

Improve bends without moving every pixel independently

Rigid pieces are appropriate for eyes, shoes and some hands. Shirt sleeves, cheeks and tails can need a small deformable mesh. Moving all pixels independently introduces enormous freedom and permits holes, folds and identity drift. Use a mesh or cage to constrain those degrees of freedom.

Skinning: attach the texture to a small set of bones

```
v(t) = sum_j w_j(v) [F_j(t) F_j(0)^(-1)] v(0)
w_j(v) >= 0, sum_j w_j(v) = 1
```

The inverse bind transform is essential: it removes the rest pose before applying the current pose. Weight normalization preserves translation behavior. Bounded biharmonic weights provide a research basis for smooth, localized influence fields [11]. Large rotations can still shrink regions under simple linear blending, so measure distortion and restrict the deformation area.

ARAP: preserve local shape while meeting joint targets

```
E = sum_(i,j) w_ij ||(v_i-v_j) - R_i(v_i0-v_j0)||^2
+ lambda_target sum_k ||v_k - target_k||^2
```

This is an illustrative ARAP-style objective: neighboring edges resist distortion while selected handles move. Exact discretization and constraints depend on the solver. Planar ARAP work demonstrates interactive shape manipulation [12]. For QH, use it locally around the elbow or shoulder rather than allowing the whole head and body to deform.

Draw the deformed texture with inverse sampling

```
p_rest = lambda1 v1_rest + lambda2 v2_rest + lambda3 v3_rest
I_output(p) = I_source(p_rest)
```

For a destination pixel in a deformed triangle, find barycentric coordinates and sample the original texture. Forward-splatting source pixels can leave holes. Resample from the original asset for every frame; repeated frame-to-frame resampling accumulates blur. Premultiplied alpha prevents hidden RGB in transparent pixels from creating edge fringes [13].

Geometry checks beyond a nice screenshot

Reject flipped triangles using signed area. Track area change, extreme stretch and contour deviation per part. Preserve seam overlaps and receiver masks. Independent part PNGs need real hidden overlap artwork; a deformation solver cannot invent a correct unseen shoulder. Subpixel anti-aliasing improves drawing, but must not alter the saved anatomical pivot.

Depth and shadows are separate inference problems

Relative depth, draw order, surface normals, material color and shading mean different things. A useful model output needs an explicit type and uncertainty. For a 2.5D mascot, correct order at overlaps can matter more than physical depth in meters.

A single image does not identify albedo and lighting uniquely

```
I(x) = A(x) * S(x)
A2(x) = k(x) A(x), S2(x) = S(x) / k(x)
```

Even this simplified diffuse model permits many decompositions for positive k. Additional priors or observations are required. Ordinal Shading and Intrinsic Image Diffusion tackle related ambiguity [14,15]. Their outputs should be treated as estimates, not measurements of the true hidden appearance.

The cartoon-specific study by Ma and colleagues [16] is especially relevant: artist-painted shading need not obey physical illumination. For Fox, dark clothing, outlines, fur details and painted shadows must not all become a detached shadow layer. This conclusion is an engineering inference from the domain distinction in that study.

Different shadow types need different ownership

Type	How it should move	Evidence needed
Painted shading on a sleeve	With the sleeve texture or its attached shading map	Material/shading annotation or confident decomposition
A hand casting a shadow on the shirt	Recomputed or artist-keyed; clipped to the shirt receiver	Caster, receiver, light and plausible depth/order
Ground/contact shadow	Owned by the floor/contact system	Foot contact, root height and art direction

An arbitrary dark overlay above every part will produce incorrect occlusions. For soft toon shadows, a controlled receiver-clipped mask can be a better practical choice than reconstructing full 3D light transport. Retain the original painted look unless a relighting operation is requested.

Depth should be optional and typed

Marigold estimates affine-invariant depth and describes single-view recovery as ill-posed [17]. Depth Anything V2 offers strong general depth estimates [18], but its scene benchmarks do not establish accuracy on Fox. Normalize relative maps only for their intended ordering purpose; do not equate a pseudo-depth value with metric distance or bone length.

Motion analysis and honest quality checks

A still image does not tell us the intended emotion, speed or action. These belong in a motion specification. A reference sequence can provide observed trajectories; a still character provides an appearance and anatomy prior. The two inputs should not be conflated.

Use tracking to estimate correspondence, not to define anatomy

RAFT predicts dense optical flow between images [19]. TAPIR tracks selected surface points across video with temporal refinement [20]. Use them as correspondence proposals for multi-frame fitting or validation. Reject occluded, low-confidence and left/right-switched points before estimating pivots. They do not supply hidden artwork or guarantee anatomically valid motion.

Continuous time; selectable output frame rate

```
u = clamp((t-t0)/(t1-t0), 0, 1)
e(u) = 10u^3 - 15u^4 + 6u^5
q(t) = q0 + e(u)(q1-q0), t_frame = n / fps
```

The easing polynomial has zero endpoint velocity and acceleration. Minimum-jerk research supports smooth hand trajectories under specific reaching assumptions [21]; a character wave still needs authored arcs, holds and secondary motion. Sampling at 24, 30 or 60 fps should not change action duration or joint trajectories. Measure speed in pixels/second and acceleration in pixels/second squared, not per frame.

A score of 95 is not 95% truth

A single weighted score can hide a detached hand, wrong mouth or changed identity. The draft currently checks contact drift, limits, clipping, loop matrices and coarse joint coverage. Joint coverage alone does not prove a connected seam, and freezing an important part can make its pixel error zero while failing the requested action.

Measure	What it tests	Proposed gate / limitation
Root + contact drift	Unintended translation; foot sliding	<= 0.5 reference-canvas px during declared contact
Seam / topology	Holes, detached limb, mesh flips	No critical gaps; no flipped triangles
Visible part IoU + boundary distance	Extraction versus approved masks	Per-part thresholds; report small parts separately
Reconstruction + perceptual change	Similarity to accepted bind/target pose	Use together; PSNR/LPIPS are not anatomy correctness [22]
Temporal residual + velocity	Jitter outside intended motion	Remove intentional motion; compare time-normalized residual
Identity + expressive success	Correct face/action and readable emotion	Visual approval on held-out poses

Use a technical score for candidate ranking only after critical gates pass. Calibrate any acceptance probability on labeled failures. Cap repair attempts; retain the best validated candidate and explain unresolved errors. Unlimited regeneration is neither reliable nor cost-efficient.

How to establish whether the upgrade is better

Test the alternatives on the same inputs. A future evaluation could use 10 characters with different proportions and clothing, with Fox reserved for the difficult cases. Annotate visible masks, joints and ordering; mark hidden content as unknown or authored. Do not score generated hidden anatomy as ground truth simply because it is plausible.

Variant	What changes	What remains fixed
A: Current authored rig	Baseline	Character source, targets, timeline and camera
B: Numeric masks + joint fit	Better extraction and registration	Runtime transforms and motion timing
C: B + local mesh deformation	Flexible bends and skinning	Same masks, targets and replacement states
D: C + pseudo-depth	Occlusion inference	Same geometry and source art
E: D + intrinsic decomposition	Optional shading/relighting	Same motion and comparison protocol

Use held-out hello, elbow bend, hand-near-face, blink, head turn, tail motion and planted-foot clips. Include overlap extremes, left/right confusion, matching-colored shirt and sleeves, thin ear tips, occluded hands, and transparent-edge artifacts. Evaluate manual correction time, memory, preparation latency, preview frame time, failure rate and human preference alongside geometry metrics.

Measure efficiency in the right place

Cache semantic embeddings, accepted masks, mesh weights and source textures per character. Factor reusable solver matrices when topology is fixed. At preview time, evaluate compact control tracks and render those assets. Full-size maps belong in preparation files; the viewport transform is only a final display mapping. SQLite can index versions and paths, while binary maps and PNGs remain files.

Proposed promotion criteria

Prefer an upgrade only if it reduces critical seam/ordering failures or correction time on held-out clips without increasing identity drift. Report median and worst cases, with sample counts and uncertainty; do not manufacture an accuracy percentage. Keep the existing rig available when model confidence is low or inference hardware is unavailable.

Draft job for a later skill update

Define a typed analysis package containing source hash, canvas units, labels, confidence maps, hidden-region provenance, rest mesh, joints, inverse bind transforms, weights, order constraints and optional appearance maps. Add adapters for candidate models and a repeatable evaluation command. Promote a model-backed default only after the ablation above. QH custom CLI integration remains a later draft job. No implementation or skill change was made for this research.

Primary references and applicability

[1] Lin et al.. See-through: Single-image Layer Decomposition for Anime Characters. SIGGRAPH 2026.
Completed semantic layers plus pseudo-depth; closest conceptual fit. Anime evidence does not validate Fox. Deformation controls and motion authoring remain separate.
Paper / author source

[2] Shitagaki Lab. See-through maintained implementation. Repository, checked 2026-10-13.
Current layer taxonomy, setup, hardware requirements and decomposition/rigging distinction. Upstream code and checkpoint terms must be tracked separately for packaging.
Paper / author source

[3] Qwen team. Qwen-Image-Layered: Towards Inherent Editability via Layer Decomposition. 2025 preprint.
General editable RGBA layer decomposition. A useful baseline; editability and reconstruction quality do not independently establish good articulation boundaries.
Paper / author source

[4] Carion et al.. SAM 3: Segment Anything with Concepts. 2025; revised 2026.
Concept-prompted detection, segmentation and tracking. Test its visible-region proposals on the mascot domain; preserve manual correction.
Paper / author source

[5] Ravi et al.. SAM 2: Segment Anything in Images and Videos. 2024 / ICLR 2025.
Promptable image/video masks with memory-based propagation. Useful for continuity of selected regions, not a complete rig.
Paper / author source

[6] Li et al.. Semantic-SAM: Segment and Recognize Anything at Any Granularity. 2023 / ECCV 2024.
Addresses semantics and mask granularity. Anatomy IDs, hidden surfaces and attachment constraints still require an application contract.
Paper / author source

[7] Chen and Zwicker. Transfer Learning for Pose Estimation of Illustrated Characters. WACV 2022.
Drawing-specific transfer learning supports using adapted joint proposals rather than treating photo-human estimates as reliable mascot anatomy.
Paper / author source

[8] Smith et al.. A Method for Animating Children’s Drawings of the Human Figure. ACM TOG 2023.
Detection, segmentation, pose estimation and mesh animation form a multi-stage pipeline. Primary implementation explanation: https://ai.meta.com/blog/using-ai-to-bring-childrens-drawings-to-life/
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[9] Poursaeed et al.. Neural Puppet: Generative Layered Cartoon Characters. WACV 2020.
Layered template deformation and appearance refinement from example images. Supports structure-aware fitting rather than unrestricted independent pixel motion.
Paper / author source

[10] Umeyama. Least-Squares Estimation of Transformation Parameters Between Two Point Patterns. IEEE TPAMI 1991.
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Paper / author source

[11] Jacobson et al.. Bounded Biharmonic Weights for Real-Time Deformation. ACM TOG 2011.
Smooth, bounded influences for points, bones and cages. Provides a grounded method for deforming textured 2D meshes.
Paper / author source

Search and access checked 13 October 2026. Most evidence is from original papers or author repositories. Reference 16 was assessed from the author abstract indexed by PubMed; no full-text findings are attributed to it. Model benchmarks in the papers are not QH benchmarks. Recommendations, equations used as engineering examples, test gates and the proposed experiment are our synthesis.

Primary references and applicability

[12] Igarashi, Moscovich and Hughes. As-Rigid-As-Possible Shape Manipulation. ACM TOG / SIGGRAPH 2005.
Interactive planar shape deformation using constraints and distortion minimization. Relevant to local sleeve/torso bending.
Paper / author source

[13] Porter and Duff. Compositing Digital Images. SIGGRAPH 1984.
Alpha representation and compositing arithmetic. Relevant to transparent edges and independent raster layers.
Paper / author source

[14] Careaga and Aksoy. Intrinsic Image Decomposition via Ordinal Shading. ACM TOG 2023.
Ordinal shading estimates support intrinsic decomposition. Illustrates why illumination analysis needs more than simple grayscale thresholding.
Paper / author source

[15] Kocsis, Sitzmann and Nießner. Intrinsic Image Diffusion for Indoor Single-view Material Estimation. CVPR 2024.
Models ambiguous material decomposition probabilistically. Indoor-photo evidence is not a validation on illustrated Fox artwork.
Paper / author source

[16] Ma et al.. Separating Shading and Reflectance From Cartoon Illustrations. IEEE TVCG 2024.
Author abstract highlights the difference between physical photo shading and artist-painted cartoon shading. DOI: 10.1109/TVCG.2023.3239364. Full publisher text was unavailable through browsing.
Paper / author source

[17] Ke et al.. Repurposing Diffusion-Based Image Generators for Monocular Depth Estimation. CVPR 2024 / Marigold.
Affine-invariant depth estimated with diffusion priors. Relative depth and unfamiliar-domain uncertainty must be explicit.
Paper / author source

[18] Yang et al.. Depth Anything V2. NeurIPS 2024.
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Paper / author source

[19] Teed and Deng. RAFT: Recurrent All-Pairs Field Transforms for Optical Flow. ECCV 2020.
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[20] Doersch et al.. TAPIR: Tracking Any Point with per-frame Initialization and temporal Refinement. ICCV 2023.
Point trajectories with temporal refinement. Provides correspondence candidates for multi-frame pivot estimation and drift checking.
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Minimum-jerk reasoning for planar reaching. An appropriate smoothness prior, not a universal prescription for expressive acting.
Paper / author source

[22] Zhang et al.. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. CVPR 2018.
Perceptual feature metrics complement simple image errors. They still do not certify anatomy, contacts, or success of a requested emotion.
Paper / author source

[23] Puhachov et al.. Generative Cutout Animation. Computer Graphics Forum 2026.
Generates rigged cutouts from skeletal motion and text using multi-pose consistency. Relevant to motion-first asset creation; different from decomposing an already accepted Fox image.
Paper / author source

Search and access checked 13 October 2026. Most evidence is from original papers or author repositories. Reference 16 was assessed from the author abstract indexed by PubMed; no full-text findings are attributed to it. Model benchmarks in the papers are not QH benchmarks. Recommendations, equations used as engineering examples, test gates and the proposed experiment are our synthesis.